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BIOLOGY

1. Different food chains interact with each other to form:
A. Food pyramid
B. Energy graph
C. Food web
D. Energy pyramid
E. Food chain complex

2. Physical basis of life is/are:
A. Nucleotide and protein
B. Nucleus
C. Cell
D. Protoplasm

3. Thought endoplasmic reticulum are involved in:
A. Protein synthesis
B. DNA synthesis
C. RNA synthesis
D. Protein modification

4. Menu ribosomes attached to endoplasmic reticulum form:
A. Abnormal function
B. Polyribosomes
C. Intense protein synthesis
D. Cell becomes malignant

5. Earthworms gaseous exchange occur through:
A. Lungs
B. Skin
C. Blood
D. Nephridia

6. Amoeba is:
A. A-cellular
B. Prokaryote
C. Multinucleated
D. None of the above

7. Nistic contains following except:
A. Cellulose
B. Ribosomes
C. Golgi apparatus
D. Chloroplasts

8. Mitochondria can be found in;
A. Virus
B. Bacteria
C. Protozoan
D. Nostoc
E. None of the above

9. Echinoderms have a brain:
A. Well developed
B. Rudimentary
C. Absent
D. None of the above

10. Male honey bees are:
A. Diploid
B. Haploid
C. Triploid
D. None

11. Spores are:
A. Diploid
B. Haploid
C. Triploid
D. None

12. Most optimum environment for the growth of fungus is:
A. Dry
B. Moist
C. Shady
D. All of the above

13. Maize endosperm is:
A. Diploid
B. Haploid
C. Triploid
D. None

14. Which of the following have an edible seed coat?
A. Guava
B. Mango
C. Pomegranate
D. Watermelon

15. Malaria is caused by:
A. Mosquito
B. Virus
C. Protozoa
D. Bacteria

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16. Which of the following is not a bacterial disease?
A. Typhoid and T.B
B. Rabies and polio
C. Diphtheria and Cholera
D. Dysentry and Diarrhea

17. How many mitotic divisions produce 12 cells:
A. 3 divisions
B. 6 divisions
C. 4 divisions
D. 12 divisions

18. Chloroplast can be traced as:
A. Root
B. Stem
C. Leaf
D. None of these

19. Guttation is due to:
A. High transpiration
B. Low relative humidity
C. Negative root pressure
D. Positive root pressure

20. When your hand touches the hot bidy, which path us taken by stimulus to withdraw your hand?
A. Sensory nuerons-> Spinal cord-> Brain-> Muscle
B. Sensory neurons-> Spinal cord-> Muscle
C. Sensory neurons-> Brain-> Spinal cord-> Motor neuron
D. Hot object-> Spinal cord-> Hand

21. A man with blood group 'A' marries a woman with blood group 'B' , child can have:
A. A & B
B. B & AB
C. A & AB
D. A, B, AB & O

22. In a population 36% people have 'Blue Eyes' ; allele frequency for blue is:
A. 36%
B. 40%
C. 60%
D. 64%

23. Which of the following is not requirementbif seed germination?
A. Oxygen
B. Water
C. Abcissic acid
D. Temperature

24. Reduction division in meiosis is:
A. Anaphase -II
B. Anaphase - I
C. Diakinesis
D. Diplotene

25. The life was evolved first:
A. On earth
B. In water
C. In sky
D. On the hill

26. The _____ structure was first proposed by Robertson:
A. Cellulo-protein
B. Star protein
C. Cyto protein
D. Lipoprotein

27. Enzymes associated with Kreb's cycle are found in :
A. Outer layer of mitochondrion
B. Inner layer of mitochondrion
C. Intermembrane space of mitochondrion
D. Matrix of mitochondrion

28. Xanthophyll is a pigment having:
A. Yellow colour
B. Red colour
C. Green colour
D. Blue colour

29. Outer membrane of mitochondrion and endoplasmic reticulum are similar in having:
A. Similar structure and constituents substances
B. The enzymes monoamine oxidase

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C. The enzyme NADH cyclochrome reductase
D. Identical metabolic function

30. There are different types of Chlorophyll like chlorophyll a and:
A. b, c, d and e
B. b
C. Xanthophyll
D. Only 'c'

CHEMISTRY

31. Which has high M.P and B.P amongst VI-A given:
A. Po
B. Te
C. Se
D. Br

32. Which is radioactive Halogen?
A. I_2
B. Cl_2
C. Br_2
D. None

33. Which has highest K.E amongst the following:
A. H2
B. O_2
C. CH4
D. All of these

34. Which is not a normal cracking phenomenon?
A. Thermal cracking
B. Catalytic cracking
C. Steam cracking
D. Normal cracking

35. What is the property of oxidizing agent?
A. Electron gainer
B. Electron donor
C. Positive charge
D. Loose Hydrogen

36. H2 + Br2 → 2HBr
The rate expression is: Rate = K[H2][Br]½ , the order of reaction is:
A. 2
B. 1.5
C. 1
D. Zero

37. Which is not a nucleotide?
A. H2O
B. H2S
C. BF3
D. NH3
E. Alil of these

38. Which alcohol can form a ketone?
A. Primary alcohol
B. Secondary alcohol
C. Tertiary alcohol
D. Methanol
E. None of the above

39. Which is not an allotrope of carbon?
A. Diamond
B. Coke
C. Methane
D. Graphite

40. What are the raw materials in the formation of urea?
A. Steam, methane, ammonia
B. Steam and ammonia
C. Methane and srteam
D. None of the above

41. In a reaction:
NO + O2 → NO3 + heat
Le-Chattlier's principle can be favoured by:
A. High pressure - high temperature
B. High pressure - low temperature
C. Low pressure - low temperature
D. Low pressure - high temperature

42. The compounds which appose enzyme activity are:
A. Co-enzymes
B. Catalysts
C. Inhibitors
D. Negative enzymes

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E. C & D

43. Decay of metal with interaction with the surrounding is called:
A. Radioactivity
B. Corrosion
C. Alloying
D. None of these

44. 18g of glucose in 1000gm of solution of glucose. The molar concentration of glucose is:
A. 1M
B. 0.1M
C. 0.02M
D. 2M

45. 1g of Hydrogen contains how many number of molecules?
A. $6.02x10^{23}$
B. $3.01x10^{23}$
C. 0
D. $1.04x10^{23}$

46. Alkanes are also known as:
A. Saturated hydrocarbons
B. Unsaturated hydrocarbons
C. Parrafins
D. None of the above

47. Isotopes differ in:
A. Proton number
B. Neutron number
C. Atomic number
D. Electron number

48. Benzene doesn't show any one if the following?
A. Oxidation
B. Addition
C. Elimination
D. Substitution

49. Noble gases have electronic configuration of:
A. ns^2, np^4
B. ns^1, np^6
C. ns^2, np^6
D. ns^2, np^3

50. Which of the following is/are electropositive element(s)
A. Alkali metals
B. Alkaline earth metals
C. Halogens
D. Both A & B

51. The ore of aluminum is:
A. Pyrite
B. Bauxite
C. Galena
D. Azurite

52. The correct sequence of higher vapour pressure is:
A. Water > Ethanol > Acetone > Ether
B. Ether > Acetone > Ethanol > Water
C. Ether > Ethanol > Acetone > Water
D. Water > Ethanol > Acetone > Ether

53. Boric acid cannot be used as:
A. Antiseptic
B. Eye washer
C. Soda bottle formula
D. Glazing tiles and ceramics

54. At constant temperature the volume of gas is $95cm^3$ at a pressure of $9.961x10^4$ N/m. What would be its volume at $10.13x10^4$ N/m^2?
A. $1cm^3$
B. $94cm^3$
C. $93cm^3$
D. $10cm^3$

55. Which one of the following is water soluble vitamin?
A. Biotin
B. Vitamin A
C. Vitamin D
D. Vitamin K

56. Honey is a mixture of _____.
A. Glucose + fructose
B. Vitamins
C. Minerals
D. All of the above

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57. The ratio of rate of diffusion of two gases is 1:3, then the ratio of their molecular weight is:
A. 1:3
B. 3:1
C. 1:9
D. 9:1

58. Which one of the following bond has the most polar character?
A. C-O
B. C-F
C. C-Br
D. C-S

59. Point the incorrect statement in case of equilibrium state:
A. It is dynamic in natre
B. It readjusts itself in the changing conditions
C. It can be attained from the side of reactants only
D. It can be attained from the either side of the reaction

60. A catalyst cannot initiate the reaction but speed up a reaction which is possible:
A. Physically
B. Thermodynamically
C. Chemically
D. In laboratory

PHYSICS

61. In which we add more impurity in transistor?
A. Base
B. Emitter
C. Collector
D. None of these

62. Special theory of relativity solves problems with:
A. Inertial frame
B. Accelerating frame
C. Both
D. None

63. Object displaced from 5m height reaches ground at a distance of 10m. What is its final velocity?
A. 10m/s
B. $10\sqrt{2}$m/s
C. 20 m/s
D. $20\sqrt{2}$ m/s

64. Temperature of a gas is directly proportional to:
A. Vibrational K.E of molecules
B. Rotational K.E of molecules
C. Angular transitional K.E of molecules
D. None

65. Gamma rays are:
A. Longitudinal waves
B. Transverse waves
C. Mechanical waves
D. None

66. In electrostatics, Newton's laws are:
A. All applicable
B. 1st law is not applicable
C. 2nd law is not applicable
D. 3rd law is not applicable

67. Vectors of same length, how they must be having angle so we get resultant twice of either?
A. 30°
B. 0°
C. 60°
D. 45°

68. When we move from pole to equator, weight of the body:
A. Increases
B. Decreases
C. Remains same
D. None of these

69. Loudness of sound is:
A. Inversely proportional to amplitude
B. Directly proportional to amplitude
C. Inversely proportional to frequency
D. Directly proportional to frequency

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70. In Newton's rings experiment, the width of air film:
A. Gradually increases
B. Gradually decreases
C. Suddenly increases
D. Suddenly decreases

71. An object falls from a bridge, at the end of 3rd second of motion its velocity is:
A. 2m/s
B. 10000mm/s
C. 3000cm/s
D. 10m/s

72. Induced emf is directly proportional to:
A. Change in emf
B. Change in inertial flux
C. Rate of change of flux
D. Both A and B
E. None

73. Faraday's law of electromagnetic induction is:
A. $\Delta\phi / \Delta t$
B. $N \Delta\phi / \Delta t$
C. $-N \Delta\phi / \Delta t$
D. $-N \Delta\phi . \Delta t$

74. AC current can be produced by:
A. Turbine
B. Motor
C. Generator
D. Transformer
E. None

75. The graph formed by Ohm's law:
A. Straight line
B. Parabola
C. Hyperbola
D. Sigmoid

76. Red cloth seems red because:
A. Absorbs red light
B. Reflects red light
C. Absorbs white light
D. Emits red light

77. 1st law of thermodynamics can be given by:
A. $\Delta E=q+\Delta W$
B. $\Delta H=E$
C. $\Delta Q=\Delta E$
D. None of the above

78. Two vectors A and B make angle θ between them, the projection of B on A is given by:
A. A. Acos θ
B. A. B / |A|
C. A. B / |B|
D. Bcos θ

79. If 42J heat is supplied to a system an work done by system is 32J. What is the change in internal energy?
A. 1J
B. 70J
C. 10J
D. 8J

80. The unit of angular acceleration is:
A. Radian/s
B. Radian^2/s^2
C. Radian/s^2
D. Radian

81. Which of the following quantities have the same dimensions?
I) Power
II) Energy
III) Torque
A. I & II
B. I & III
C. II & III
D. I, II & III

82. If A.B=0, we conclude that:
A. Either of two vectors is a null vector
B. Both of the vectors are null
C. The vectors are mutually perpendicular
D. All of the above

83. The scalar product of two vectors A and B at an angle θ with each other is :
A. ABsin θ
B. AB

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C. ABcos θ
D. ABsec θ

84. The cross product of a vectors A with itself (A X A) is equal to:
A. A
B. 1
C. 2A
D. Zero

85. Which one of the following process obeys Boyle's law?
A. Isothermal
B. Isobaric
C. Adiabatic
D. Non-adiabatic
E. Alil of the above

86. Charge on each plate of a parallel plate capacitor is 6μC and capacitance is 4f. What is the potential difference between plates?
A. 2.5 Coulomb
B. 3.8 Farad
C. 1.8 Volt
D. 1.5 Volt
E. 3.5 Joule

87. Lenz's law is accordance with law of conservation of:
A. Momentum
B. Charge
C. Angular momentum
D. Energy
E. Mass

88. Which of the following purpose is served by the transformer?
A. Converts AC into DC
B. Converts DC into AC
C. Changes the voltage of AC
D. Decreases the power supply
E. Increases the power supply

89. Which of the following statements is incorrect in perspective of modern physics?
A. Energy has inertia
B. Time dilation phenomenon does not occur in daily life
C. Light is electromagnetic radiation
D. Universe is continuously contracting

90. Visible spectrum of light lies in:
A. Paschen Series
B. Balmer Series
C. Lyman Series
D. Bracket Series
E. Pfund Series

ENGLISH

91. LOATHSOME (Synonym)
A. Handsome
B. Unpleasant
C. Brilliant
D. Compare

92. COMPRISE (synonym)
A. Consists of
B. Forms
C. To agree
D. Accumulate

93. BLISS (antonym)
A. Happiness
B. Misery
C. Comfort
D. Pleasure

94. EXHIBIT (antonym)
A. Conceal
B. Showy
C. To sell
D. Conceive

95. 'Inflict pain' to other means:
A. To injure
B. To harness
C. Feeling pain
D. Giving pain

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ANSWER KEY

Q NO	ANS KEY	Q NO	ANS KEY	Q NO	ANS KEY	NO	ANS KEY
1	C	26	A	51	B	76	B
2	C	27	D	52	A	77	D
3	A	28	A	53	C	78	B
4	B	29	A	54	C	79	C
5	B	30	B	55	A	80	C
6	D	31	A	56	A	81	C
7	C	32	D	57	D	82	D
8	C	33	D	58	B	83	C
9	C	34	D	59	C	84	D
10	A	35	A	60	C	85	A
11	A	36	B	61	B	86	D
12	B	37	C	62	A	87	D
13	A	38	B	63	B	88	C
14	A	39	C	64	D	89	D
15	C	40	D	65	B	90	B
16	B	41	B	66	A	91	B
17	C	42	C	67	B	92	A
18	C	43	B	68	C	93	B
19	D	44	B	69	D	94	A
20	B	45	B	70	A	95	D
21	D	46	A	71	C		
22	A	47	B	72	C		
23	C	48	B	73	C		
24	B	49	C	74	C		
25	B	50	D	75	A		

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